Computer Graphics Transformations

Lecture 5

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Outline

- 2D Transformations
  - Scaling
  - Rotation
  - Translation
  - Reflections
  - Shearing
- Homogeneous Coordinates
- Inverse transformations
- Composite transformations
- 3D Transformations
  - Translation
  - Rotation
  - Scaling

Basic 2-D Transformations

- “ Transformations are the operations applied to geometrical description of an object to change its position, orientation, or size are called geometric transformations”
- Various Transformation techniques :
  - Translation
  - Rotation
  - Scaling
  - Reflection
  - Shearing

Scaling

- Scaling is the process of expanding or compressing the dimensions of an object.
- Alters size of an object.
- Positive Scaling Constants S_x and S_y are used to describe the changes in lengths with respect to *x* and *y* directions respectively

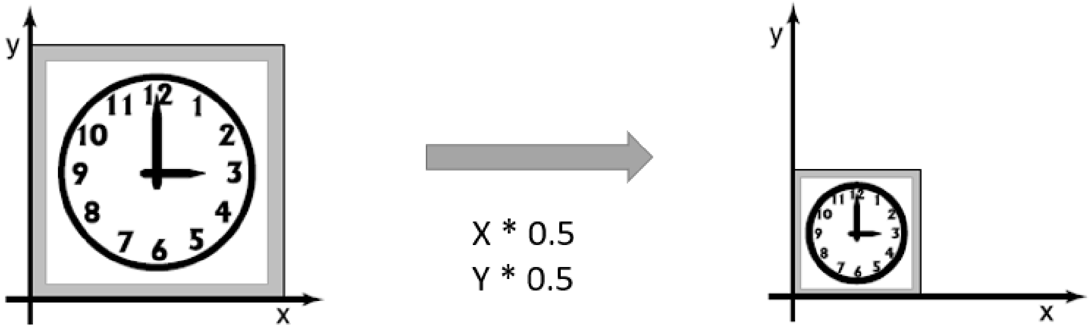
$$X' = X * S_x$$
$$Y' = Y * S_y$$

- *Homogeneous/Uniform scaling* – If Both the scaling constants have same vale S , scaling transformation is said to be *'Uniform'*
- *Non-Uniform* – Scaling Constants having different values
- If S > 1, it is a *magnification*

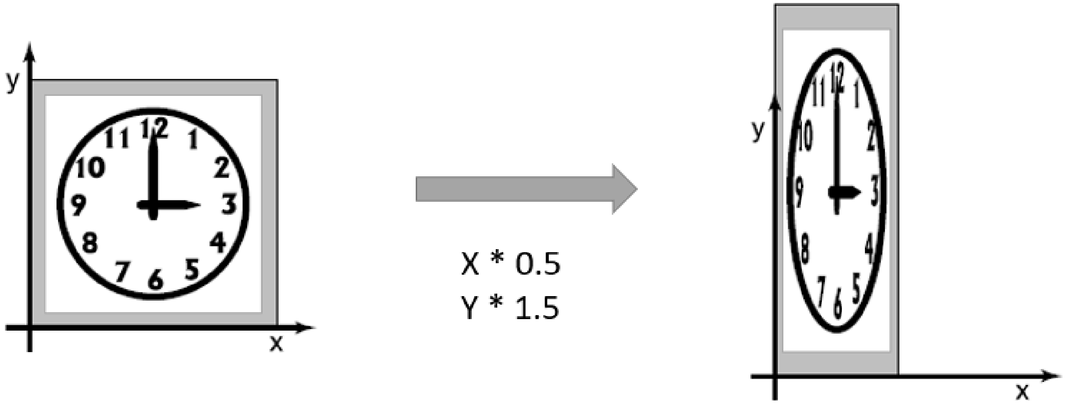
  And S < 1 , it is a *reduction*

Scaling

Uniform Scaling



Non-Uniform Scaling



Scaling

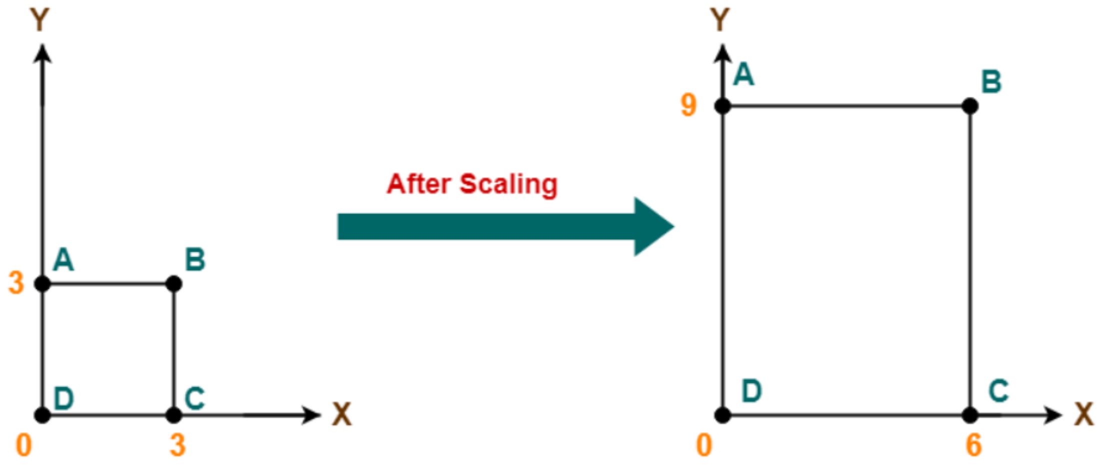
- Scaling Operation : $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} S_x x \\ S_y y \end{bmatrix}$

- Matrix Form : $\begin{bmatrix} x' \\ y' \end{bmatrix} = \underbrace{\begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix}}_{\text{Scaling matrix}} \begin{bmatrix} x \\ y \end{bmatrix}$

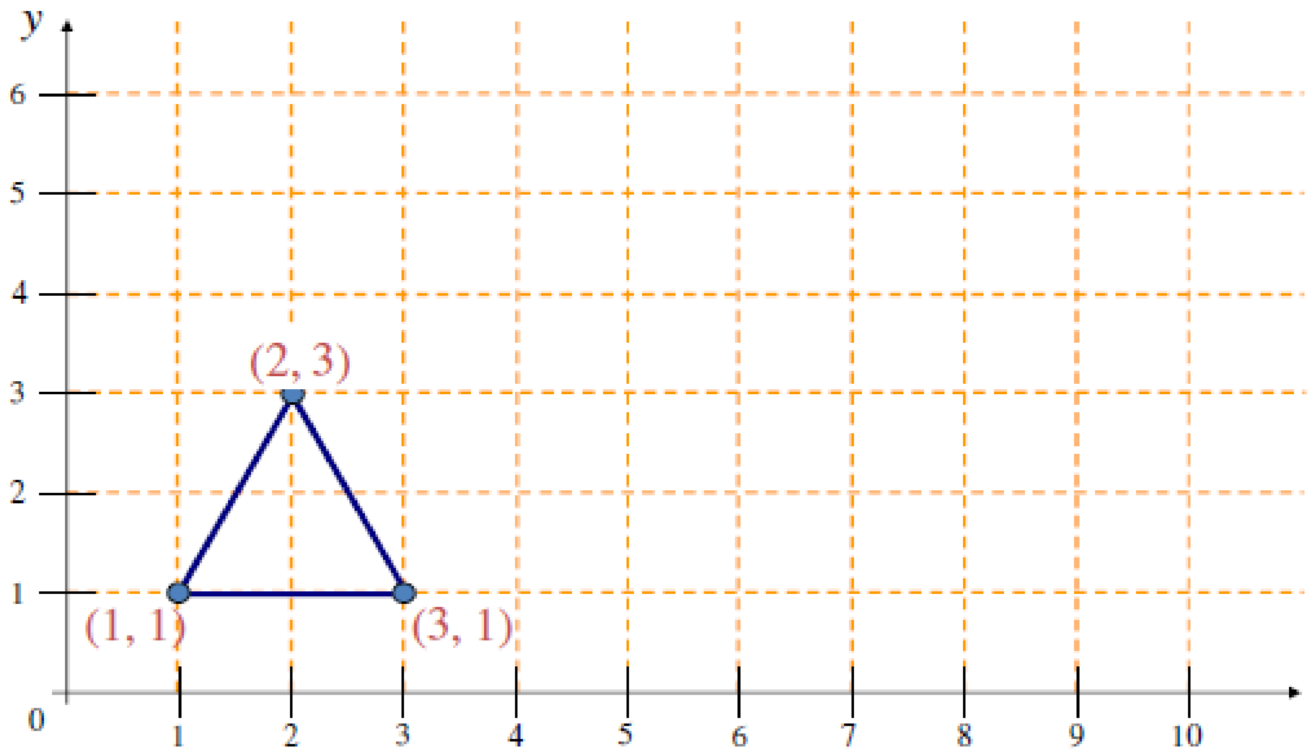
Example :

Given a square object with coordinate points A(0, 3), B(3, 3), C(3, 0), D(0, 0). Apply the scaling parameter 2 towards X axis and 3 towards Y axis and obtain the new coordinates of the object.

- Solution :
  - Old corner coordinates of the square = A (0, 3), B(3, 3), C(3, 0), D(0, 0)
  - Scaling factor along X axis = 2
  - Scaling factor along Y axis = 3
  - <u>For Coordinates A(0, 3)</u>
    - $X_{new} = X_{old} \times S_x = 0 \times 2 = 0$
    - $Y_{new} = Y_{old} \times S_y = 3 \times 3 = 9$
    - → New coordinates of corner A after scaling = (0, 9).
  - <u>For Coordinates B(3, 3)</u>
    - $X_{new} = X_{old} \times S_x = 3 \times 2 = 6$
    - $Y_{new} = Y_{old} \times S_y = 3 \times 3 = 9$
    - → New coordinates of corner B after scaling = (6, 9)
  - <u>For Coordinates C(3, 0)</u>
    - $X_{new} = X_{old} \times S_x = 3 \times 2 = 6$
    - $Y_{new} = Y_{old} \times S_y = 0 \times 3 = 0$
    - → New coordinates of corner C after scaling = (6, 0)
  - <u>For Coordinates D(0, 0)</u>
    - $X_{new} = X_{old} \times S_x = 0 \times 2 = 0$
    - $Y_{new} = Y_{old} \times S_y = 0 \times 3 = 0$
    - → New coordinates of corner D after scaling = (0, 0)



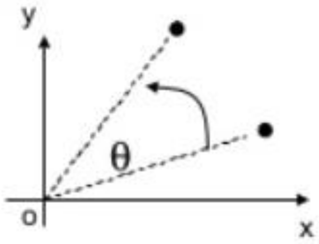
- Example : Scale by (2, 2)



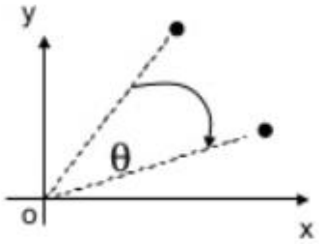
- **Watch Out** : Object grow and move

Rotation

- Rotation can be defined as repositioning of an object on a circular path from one angle to another.
- Object rotates θ^0 about origin.
- Default rotation center : Origin (0,0)



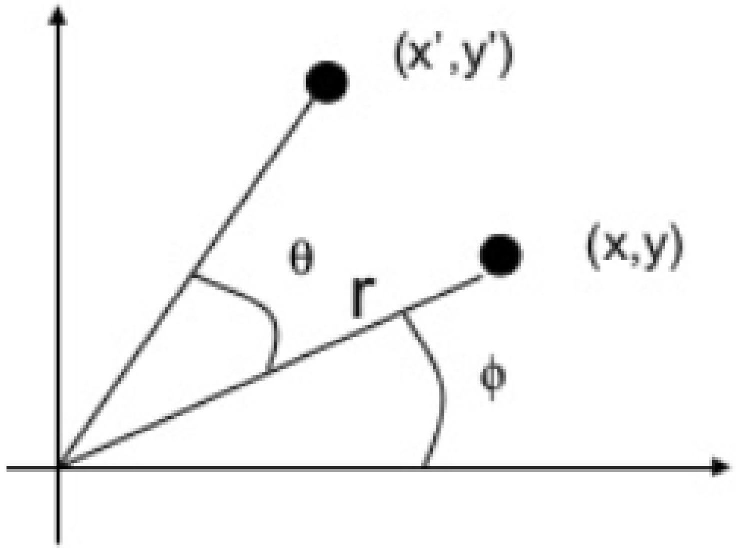
$\theta > 0$: Rotate counter clockwise



$\theta < 0$: Rotate clockwise

Rotation

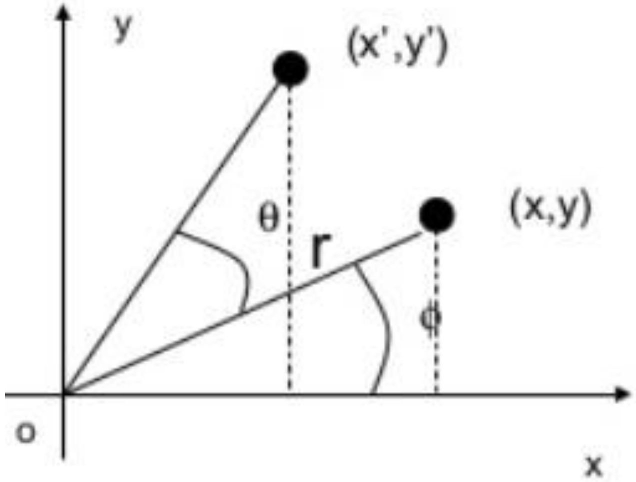
- (x,y) -> Rotate about the origin by θ , → $(x'y')$

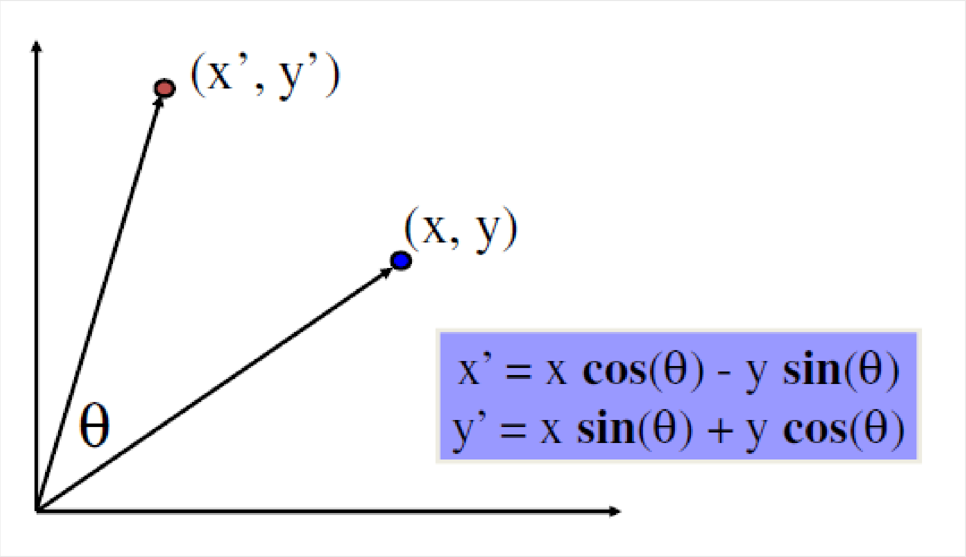


- How to compute $(x'y')$?

Rotation

- $X = r \cos \phi$ $y = r \sin \phi$
- $X' = r.\cos(\phi + \theta)$ $y' = r.\sin(\phi + \theta)$



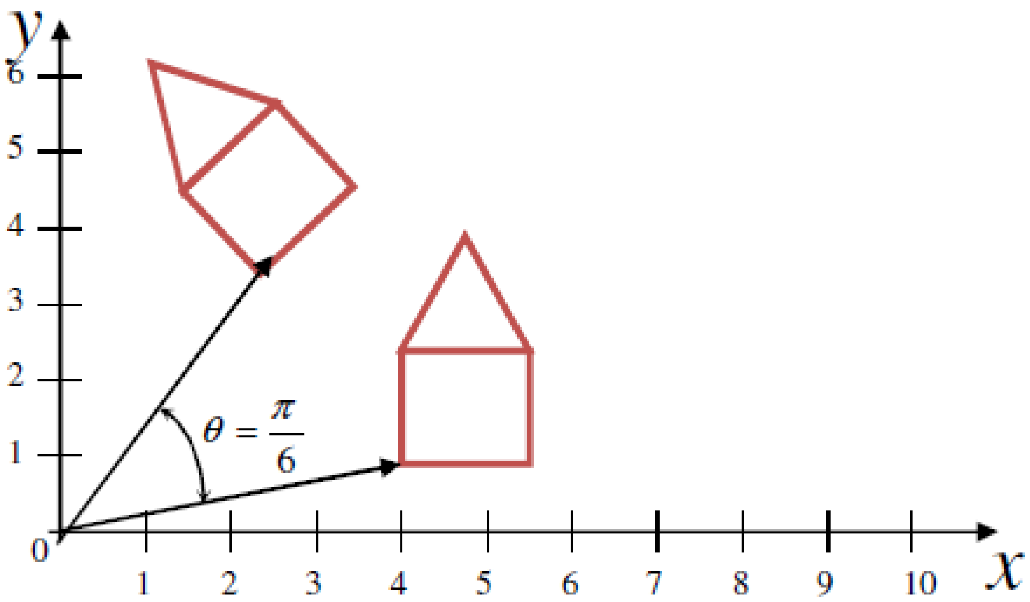


- Matrix form :

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Rotation

- Rotate all the coordinates by a specified angle
  - $x' = x\cos\theta - y\sin\theta$
  - $y' = x\sin\theta + y\cos\theta$
- Points are always rotated about the origin



Example :

A point lies on (4,4) subjected to 30^0 rotation counterclockwise direction. Find the new coordinate of the point ?

- Solution :
  - Old coordinates of the point = (X_{old}, Y_{old}) = (4, 4)
  - Rotation angle = θ = 30^0
  - Applying the rotation equations,
    - $X_{new} = X_{old} \times \cos\theta - Y_{old} \times \sin\theta$

      $= 4 \times \cos 30^\circ - 4 \times \sin 30^\circ$

      $= 4 \times (\sqrt{3} / 2) - 4 \times (1 / 2)$

      $= 2\sqrt{3} - 2$

      $= 2(\sqrt{3} - 1)$

      $= 2(1.73 - 1)$

      $= 1.46$

    - $Y_{new} = X_{old} \times \sin\theta + Y_{old} \times \cos\theta$

      $= 4 \times \sin 30^\circ + 4 \times \cos 30^\circ$

      $= 4 \times (1 / 2) + 4 \times (\sqrt{3} / 2)$

      $= 2 + 2\sqrt{3}$

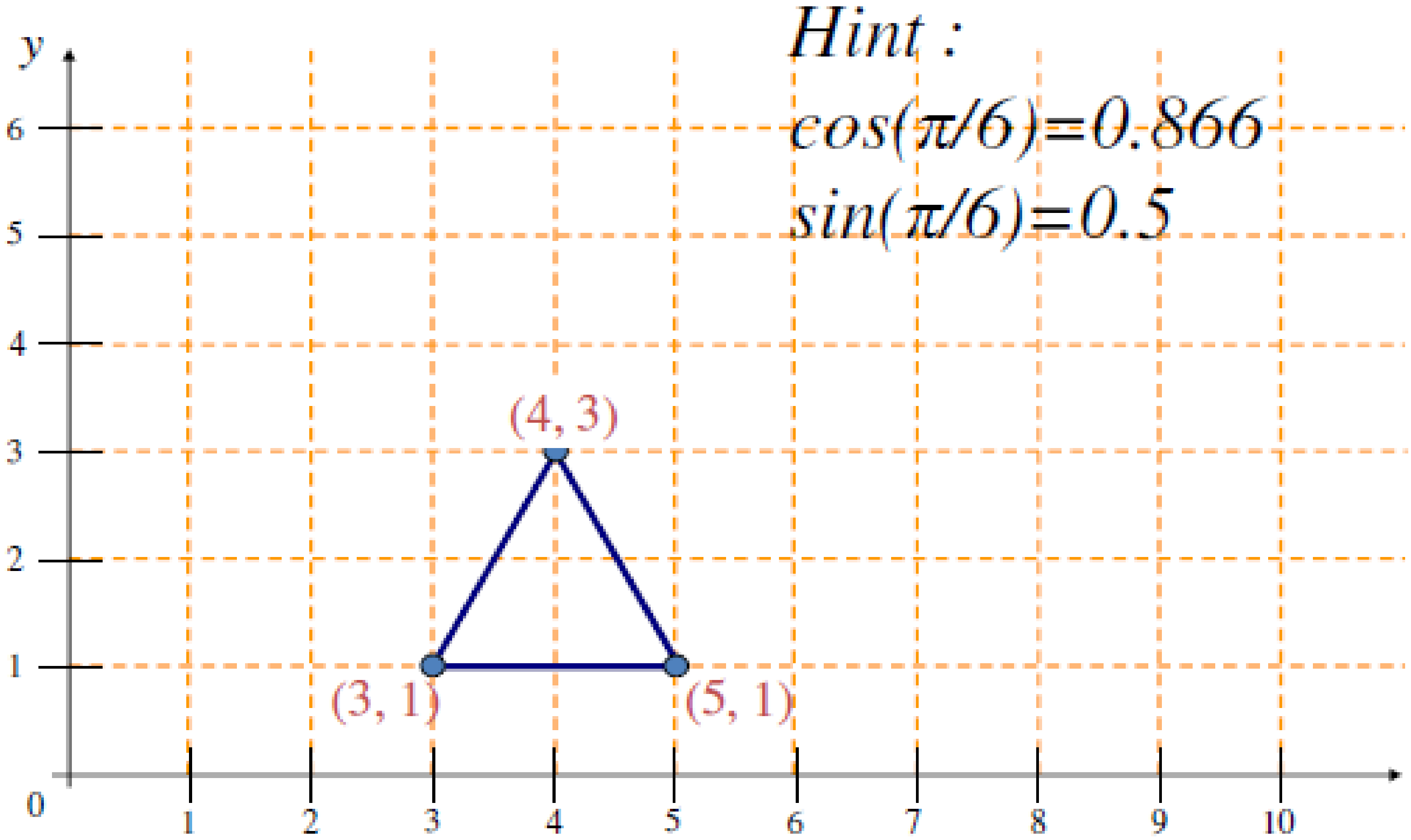
      $= 2(1 + \sqrt{3})$

      $= 2(1 + 1.73)$

      $= 5.46$

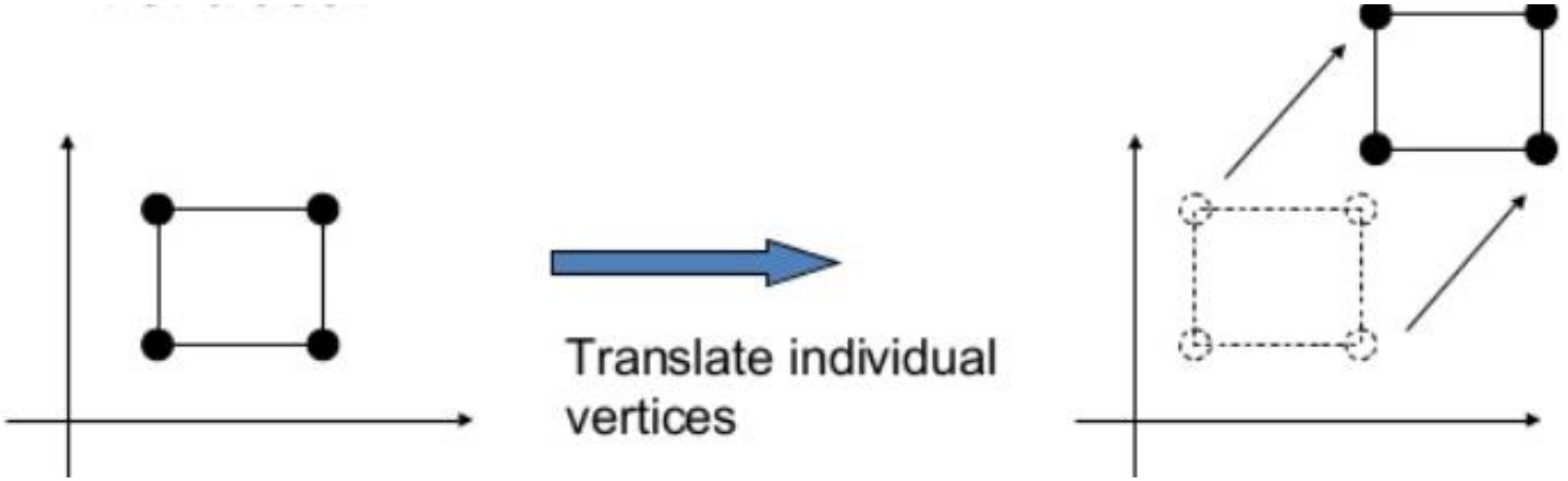
New Coordinate (1.46, 5.46)

- Example : rotate by π/6



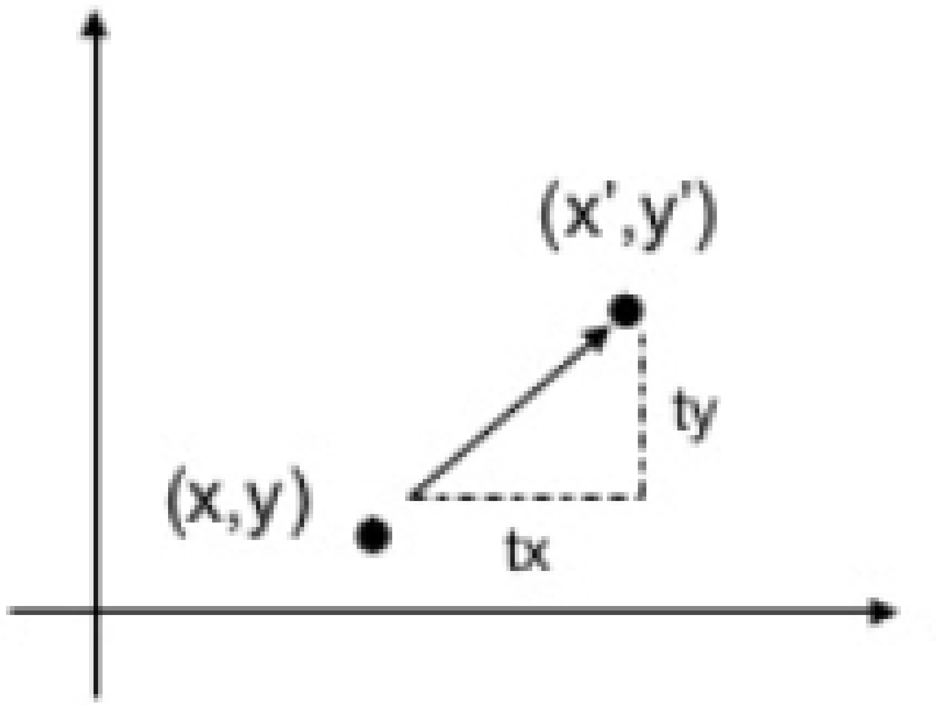
Translation

- In translation, an object is displaced a given distance and a direction from its original position.
- Translation is always considered as straight path
- A translation moves all points in an object along the same straight-line path to new positions.



Translation

- Given a point (x,y), and the translation distance (tx,ty)
- The new point (X' , Y') :
  - X' = X + Tx
  - Y' = Y + Ty
- Or in Matrix Form :
  - $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} Tx \\ Ty \end{bmatrix}$



Example :

Given a point with coordinate points P(2, 2). Apply the translation with distance 6 towards X axis and 4 towards Y axis. Obtain the new coordinates of the point.

- Solution:

X' = X + Tx

= 2 + 6

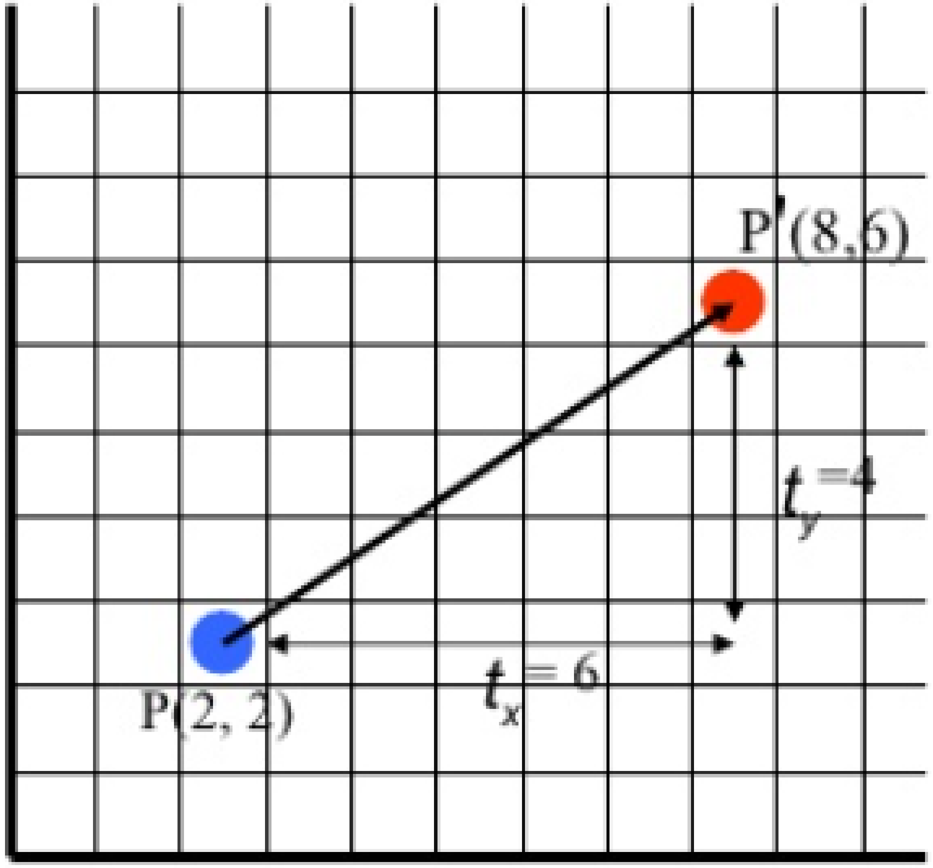
= 8

Y' = Y + Ty

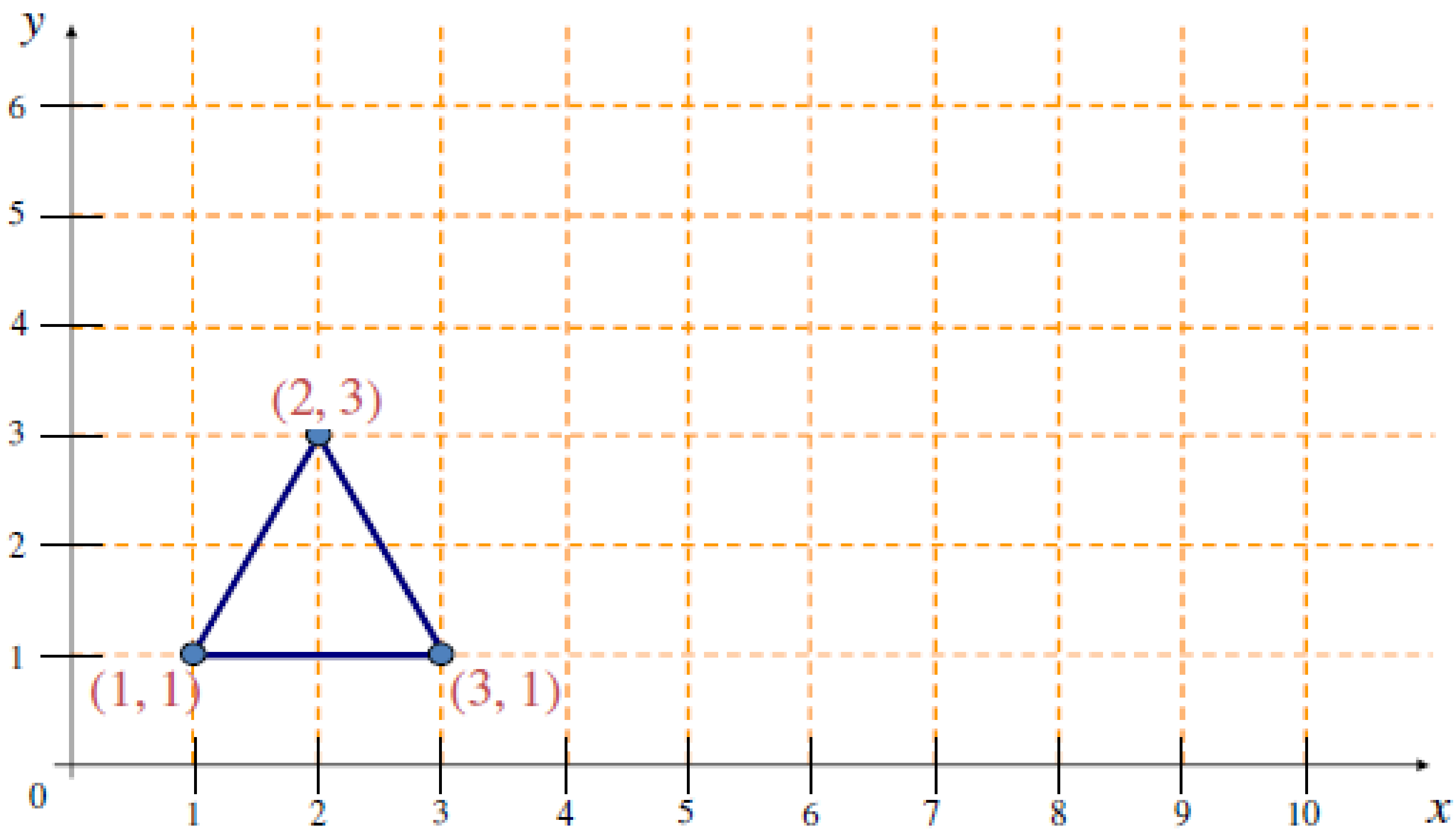
= 2 + 4

= 6

New Coordinate → (8,6)



- Example : Translate by (5, 3)



Homogeneous Coordinates

- A point *(x, y)* can be re-written in homogeneous coordinates as *(xh, yh, h)*
- The homogeneous parameter *h* is a nonzero value such that:

$$x = \frac{x_h}{h} \qquad y = \frac{y_h}{h}$$

- We can then write any point *(x, y)* as *(hx, hy, h)*
- We can conveniently choose *h = 1* so that *(x, y)* becomes *(x, y, 1)*

Why Homogeneous Coordinates?

- Mathematicians commonly use homogeneous coordinates as they allow scaling factors to be removed from equations
- We will see in a moment that all of the transformations we discussed previously can be represented as 3*3 matrices
- Using homogeneous coordinates allows us use matrix multiplication to calculate transformations – extremely efficient!

Homogeneous Translations/Rotation/Scaling

- Translation –
  - The translation of a point by *(dx, dy)* can be written in matrix form as:
  - $\begin{bmatrix} 1 & 0 & \mathrm{d}x \\ 0 & 1 & \mathrm{d}y \\ 0 & 0 & 1 \end{bmatrix}$ x $\begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$ = $\begin{bmatrix} 1*x & + & 0*\mathrm{y} & + & \mathrm{dx}\ *1 \\ 0*\mathrm{x} & + & 1*y & + & \mathrm{d}y*1 \\ 0*\mathrm{x} & + & 0*y & + & 1*1 \end{bmatrix}$ = $\begin{bmatrix} d\,x+x \\ dy+y \\ 1 \end{bmatrix}$
- Scaling –
  - $\begin{bmatrix} \mathrm{Sx} & 0 & 0 \\ 0 & \mathrm{Sy} & 0 \\ 0 & 0 & 1 \end{bmatrix}$ x $\begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$ = $\begin{bmatrix} \mathrm{Sx}*x & + & 0*\mathrm{y} & + & 0\ *1 \\ 0*\mathrm{x} & + & \mathrm{Sy}*y & + & 0*1 \\ 0*\mathrm{x} & + & 0*y & + & 1*1 \end{bmatrix}$ = $\begin{bmatrix} Sx\ *x \\ Sy*y \\ 1 \end{bmatrix}$
- Rotation-
  - $\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$ x $\begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$ = $\begin{bmatrix} \cos\theta*x & -\sin\theta*\mathrm{y} & + & 0\ *1 \\ \sin\theta*\mathrm{x} & +\cos\theta*y & + & 0*1 \\ 0*\mathrm{x} & +\ 0*y & + & 1*1 \end{bmatrix}$ = $\begin{bmatrix} \cos\theta*x-\sin\theta*\mathrm{y} \\ \sin\theta*\mathrm{x}+\cos\theta*y \\ 1 \end{bmatrix}$

Inverse Transformations

- Opposite Transformations
- Transformations can easily be reversed using inverse transformations

- Inverse Translation Matrix $T^{-1} = \begin{bmatrix} 1 & 0 & -\mathrm{d}x \\ 0 & 1 & -\mathrm{d}y \\ 0 & 0 & 1 \end{bmatrix}$ Inverse Scaling Matrix $S^{-1} = \begin{bmatrix} \frac{1}{S_x} & 0 & 0 \\ 0 & \frac{1}{S_y} & 0 \\ 0 & 0 & 1 \end{bmatrix}$

- Inverse Rotation Matrix $R^{-1} = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Composite Transformation

- No of transformations that are combined into single one called as composition.

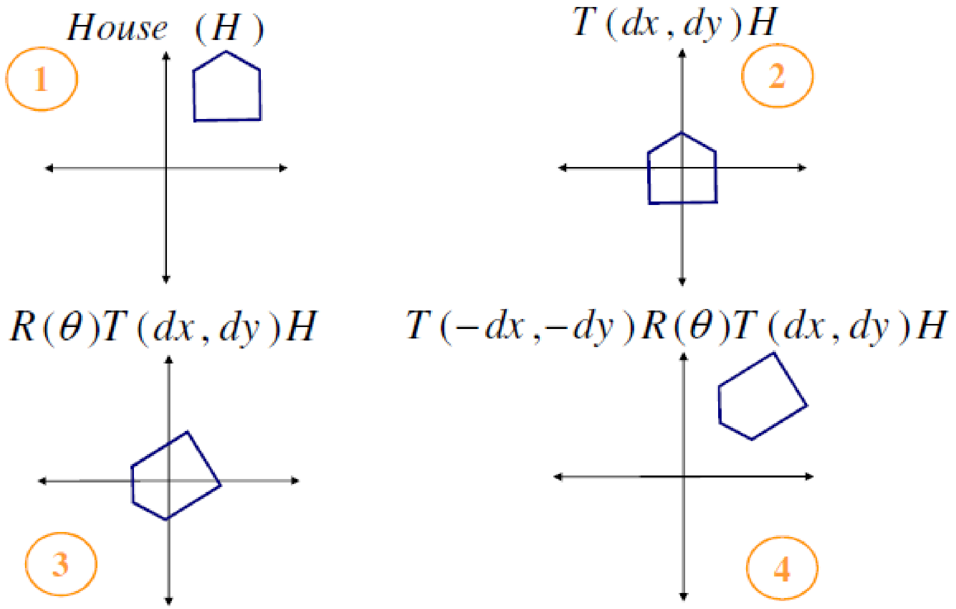
  e.g : Perform a rotation of an arbitrary point

  Can be perform it by sequence of three transformations

  -Translation

  - Rotation

  - Inverse Translation



Composite Transformation

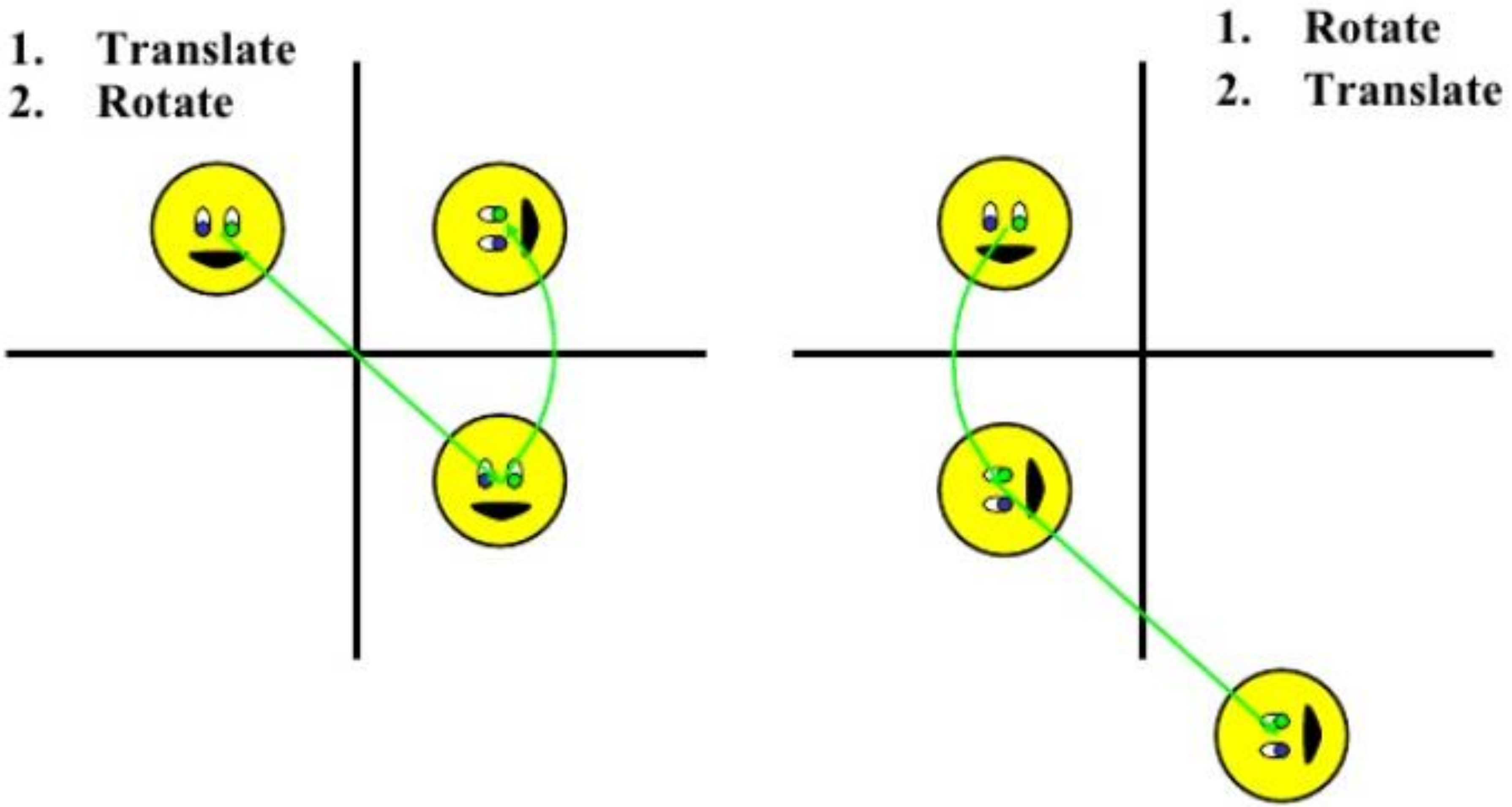
- The three transformation matrices are combined as follows
- $\begin{bmatrix} 1 & 0 & \mathrm{d}x \\ 0 & 1 & \mathrm{d}y \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 1 & 0 & -\mathrm{d}x \\ 0 & 1 & -\mathrm{d}y \\ 0 & 0 & 1 \end{bmatrix}$
- *T* (*dx*, *dy*). *R*(q). *T* (*–dx*,*–dy*)
- **Important :**Matrix multiplication is not commutative so order matters

Composite Transformation

- Matrix multiplication is associative
  - M3 x M2 x M1 = (M3 x M2) x M1 = M3 x (M2 x M1)
- Transformation products may not be commutative A x B != B x A
- Some cases where A x B = B x A

A	**B**
translation	translation
scaling	scaling
rotation	rotation
uniform scaling	rotation

Order Matters..



Exercise:

Rotate a triangle with vertices (10,20), (10,10), (20,10) about the origin by 30 degrees and then translate it by $t_x=5$, $t_y=10$

- Solution :

We compute the rotation matrix:

$$B = \begin{bmatrix} \cos 30 & -\sin 30 & 0 \\ \sin 30 & \cos 30 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

And we compute the translation matrix:

$$A = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 10 \\ 0 & 0 & 1 \end{bmatrix}$$

Then, we compute M=A·B

$$M = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 10 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$M = \begin{bmatrix} 1*0.866+0*0.5+5*0 & 1*-0.5+0*0.866+5*0 & 1*0+0*0+5*1 \\ 0*0.866+1*0.5+10*0 & 0*-0.5+1*0.866+10*0 & 0*0+1*0+10*1 \\ 0*0.866+0*0.5+1*0 & 0*-0.5+0*0.866+1*0 & 0*0+0*0+1*1 \end{bmatrix}$$

$$M = \begin{bmatrix} 0.866 & -0.5 & 5 \\ 0.5 & 0.866 & 10 \\ 0 & 0 & 1 \end{bmatrix}$$

Then, we compute the transformations of the 3 vertices:
Transformation of vertex (10,20):

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 0.866 & -0.5 & 5 \\ 0.5 & 0.866 & 10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 20 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.866*10+(-0.5)*20+5*1 \\ 0.5*10+0.866*20+10*1 \\ 0*10+0*20+1*1 \end{bmatrix} = \begin{bmatrix} 3.66 \\ 32.32 \\ 1 \end{bmatrix}$$

Transformation of vertex (10,10):

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 0.866 & -0.5 & 5 \\ 0.5 & 0.866 & 10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 10 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.866*10+(-0.5)*10+5*1 \\ 0.5*10+0.866*10+10*1 \\ 0*10+0*10+1*1 \end{bmatrix} = \begin{bmatrix} 8.66 \\ 23.66 \\ 1 \end{bmatrix}$$

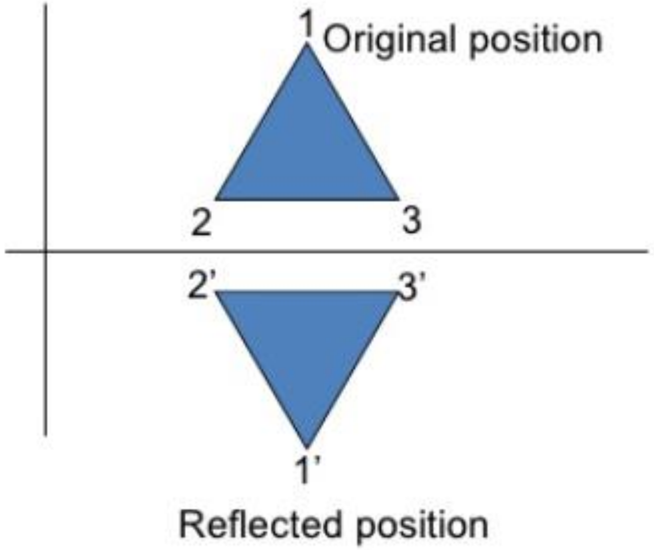
Transformation of vertex (20,10):

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 0.866 & -0.5 & 5 \\ 0.5 & 0.866 & 10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 20 \\ 10 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.866*20+(-0.5)*10+5*1 \\ 0.5*20+0.866*10+10*1 \\ 0*20+0*10+1*1 \end{bmatrix} = \begin{bmatrix} 17.32 \\ 28.66 \\ 1 \end{bmatrix}$$

The resultant coordinates of the triangle vertices are (3.66,32.32), (8.66,23.66), and (17.32,28.66) respectively.

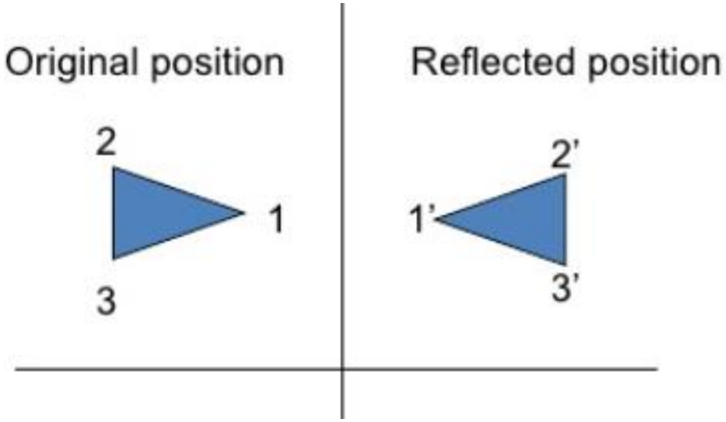
Reflection

- Reflection : is a transformation that produces a mirror image of an object. It is obtained by rotating the object by 180^0 about the reflection axis.



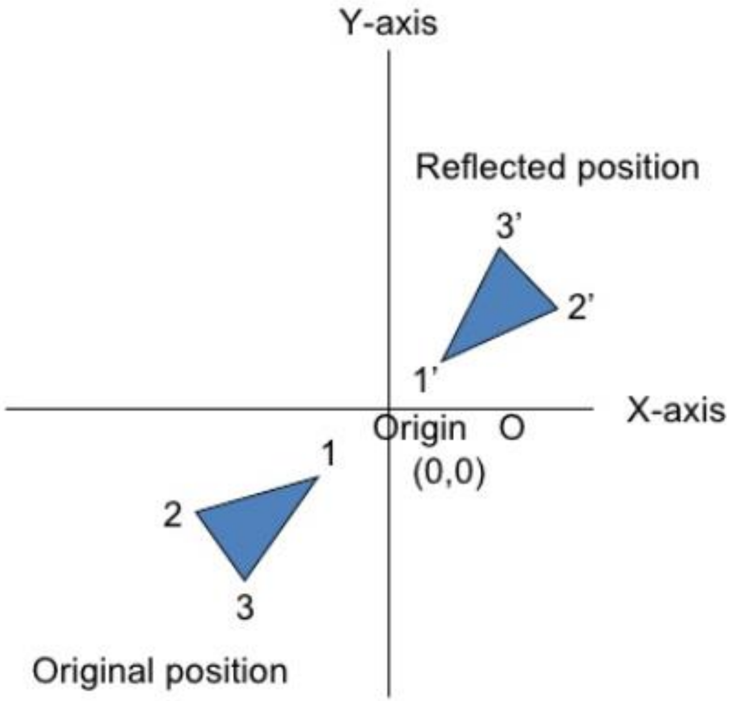
Reflection about the line y=0, the X- axis , is accomplished with the transformation matrix :

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



Reflection about the line x=0, the Y- axis , is accomplished with the transformation matrix:

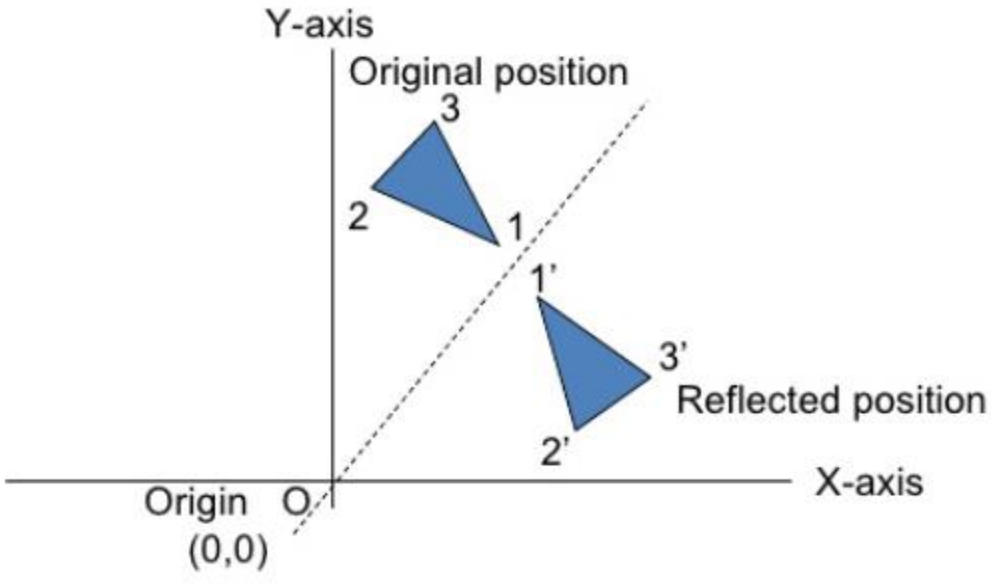
$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



Reflection of an object relative to an axis perpendicular to the xy plane and passing through the coordinate origin. matrix of this transformation is:

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

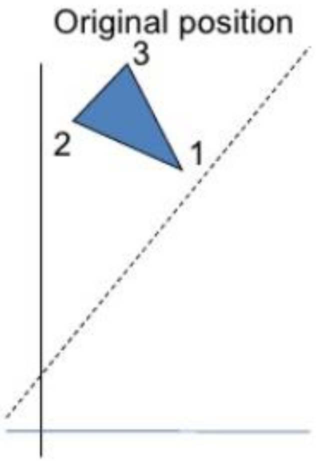
Here both x and y values will be reversed.



Reflection of an object with respect to the straight line where x = y,
The transformation matrix would be :

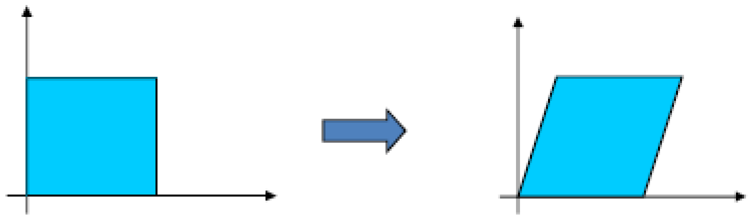
$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Q. How to get the Reflection of an arbitrary axis where y = mx + b.



Shearing

- Shear is a transformation that distorts the shape of an object such that the transformed shape appears as if the object were composed of internal layers that had been caused to slide over each other
- Two common shearing transformations are those that shift coordinate x values and those that shift y values



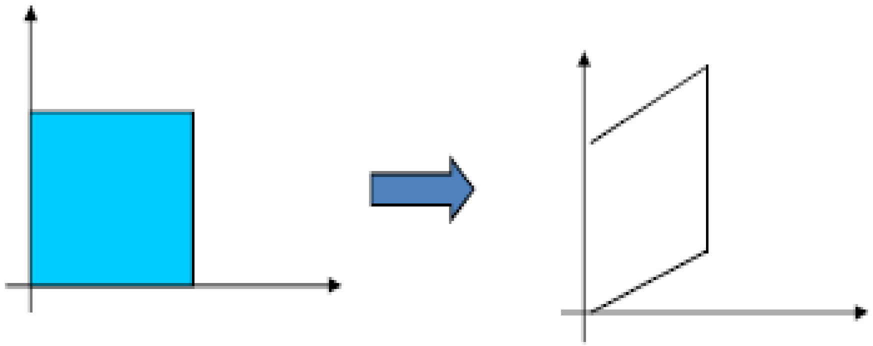
Y coordinates are unaffected, but x coordinates are translated linearly with y.

That is:

- $y' = y$
- $x' = x + y * shx$

Matrix :

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & \mathrm{sh}\,x & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$



X coordinates are unaffected, but y coordinates are
translated linearly with X,
That is :
- $x' = x$
- $y' = y + x * shy$

Matrix would be ,

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \text{sh}\,y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

- Shearing Transformations :

  - Shearing will not change the area of the object

  - Any 2D shearing can be done by a rotation, followed by a scaling, and followed by a rotation

3D Transformations

- Transformation that takes place in a 3D plan is known as '3D Transformation'.
- Methods for object modeling transformation in three dimensions are extended from two dimensional methods by including consideration for the z coordinate.
- Homogeneous coordinates : have 4 components
- Transformation matrix : 4 x 4 matrix

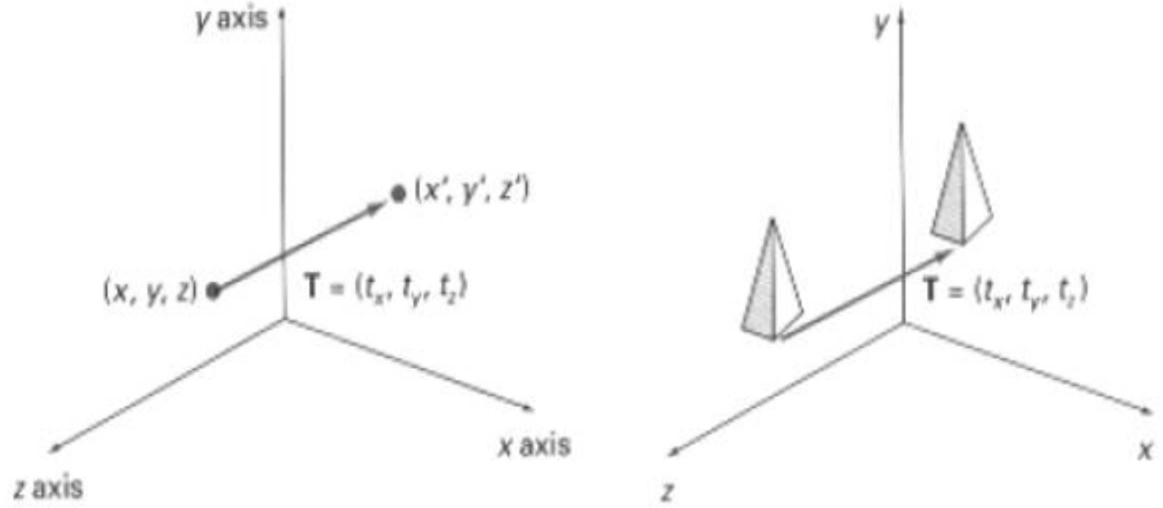
$$\begin{bmatrix} a & b & c & tx \\ d & \mathrm{e} & f & t_y \\ 9 & h & j & tz \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

3D Translation

- Movement of an object from one position to another position in 3D plane is 3D Translation
- In 3-dimensional homogeneous coordinate representation , a point is transformed from position P = (x, y , z) to P'=(x', y', z')

P = T . P

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & tx \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & tz \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

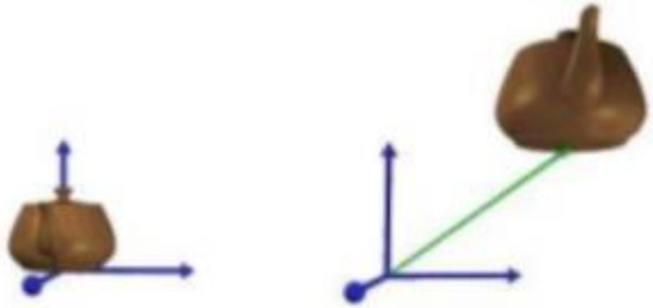


3D Translation

- The matrix representation is equivalent to the three equation.

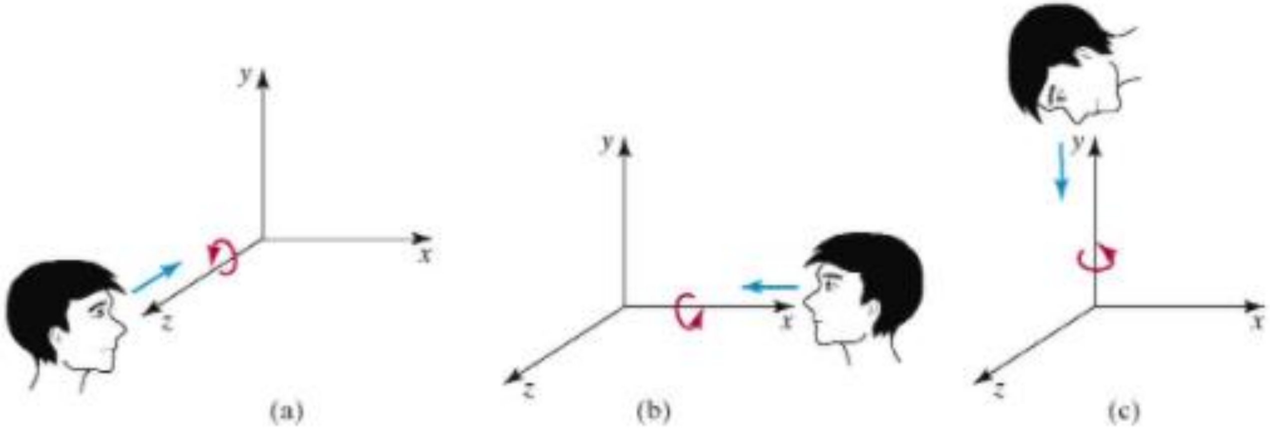
  $x' = x + t_x$, $y' = y + t_y$, $z' = z + t_z$
- Where parameter tx, ty, tz are specifying translation distance for the x , y , z coordinate direction and x , y , z are assigned any real value.

- Translate an object by translating each vertex of the object



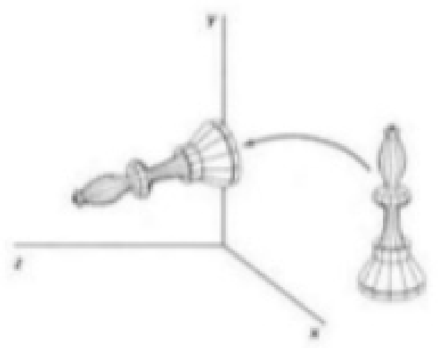
3D Rotation

- Rotation of an object in space is done by means of rotation around coordinate axes
- Coordinate axes rotation
  - X-axis rotation
  - Y-axis rotation
  - Z-axis rotation

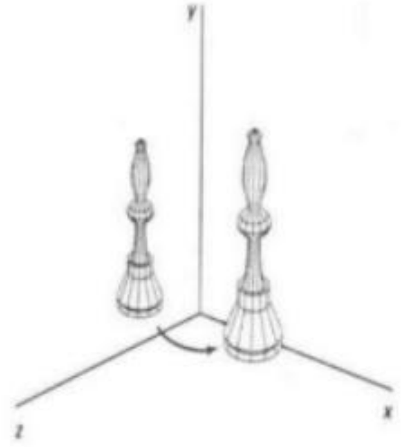


  - Positive rotations about a coordinate axis are counterclockwise, when looking along the positive half of the axis toward the origin

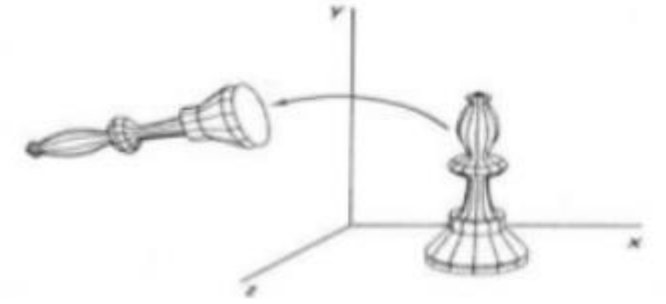
- About X-axis :
  - X' = x y' = y cos θ - z sin θ Z' = y sin θ + z cos θ
  - $$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



- About Y-axis :
  - X' = x cos θ + z sin θ Y' = y Z' = z cos θ - x sin θ
  - $$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & \sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



- About Z-axis :
  - X’ = x $\cos\theta$ - y $\sin\theta$ Y’ = x $\sin\theta$ $-$ y $\cos\theta$ Z’=z
  - $$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



? How to get the rotation about an axis parallel to one of the coordinate Axis?

In special cases where an object is to be rotated about an axis that is parallel to one of the coordinate axis, we can obtain the desired rotation with the following transformation sequence.

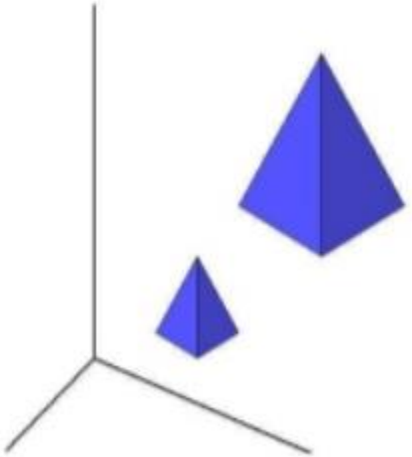
1. Translate the object so that the rotation axis coincides with the parallel coordinate axis (for simplicity, let us take x-axis).
2. Perform the specified rotation about that axis.
3. Translate the object so that the rotation axis is moved back to its original position.

3D Scaling

- Changes the size of the object and repositions the object relative to the coordinate origin.
- Either expand or compress the dimensions of the object .
- Scaling can be achieved by multiplying the original coordinates of the object with scaling factor to get the desired result.

X' = Sx. X Y'= Sy. Y Z'=Sz. Z

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & o & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



Questions?